

*Sem 5*

2<sup>nd</sup> INTERNAL/ SOFT COMPUTING/ PEC-CS501B / 5<sup>TH</sup> SEM-CSE/DEC-2024/ FM: 15

1. With a neat flowchart, explain the simple form of Genetic Algorithm. 8
2. What is the difference between Fuzzyness vs. Probability? 3
3. Show how individuals are selected from initial population by 'Sroulette wheel selection' method. 4

*P. Gita*

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2<sup>nd</sup> Class Test, Discrete Mathematics 2024 Paper code PCC-CS502. F.M 15

1. Solve the recurrence relation  $a_{n+2} - 4a_{n+1} + a_n = 3^n$ ,  $n \geq 0$ , with  $a_0 = 2$  and  $a_1 = 1$  using generating function
2. Find the total number of integers between 1 and 3000 that are not divisible by any of the integers 2, 3, 7,  $2x_n \vee n$
3. Show that  $((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$  is a tautology.

Write

**DBMS (Class Test II/ PCC-CS503)**  
**Department of Computer Science and Engineering**  
**2024**

**Full marks: 15**

**Time: 45 min**

**Answer any three**

- |                                                                                |   |
|--------------------------------------------------------------------------------|---|
| 1) What is transaction? Describe ACID property of a transaction with examples. | 5 |
| 2) What is schedule? Describe recoverable and cascade less schedule.           | 5 |
| 3) Explain conflict serializable schedule with one example.                    | 5 |
| 4) What is two-phase locking protocol? How does it guarantee serializability?  | 5 |

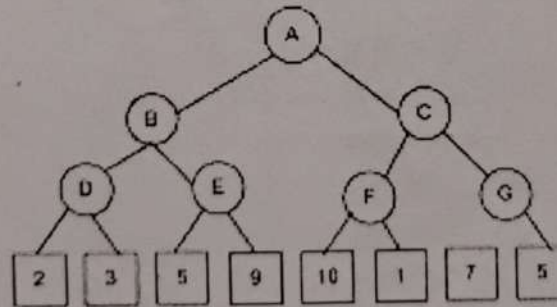
**2<sup>nd</sup> INTERNAL/ OPERATING SYSTEM/ PCC-CS504 / 5<sup>TH</sup> SEM-CSE/DEC-2024/ FM: 15**

1. Why page size is always a power of two? What is demand paging? (2+3)
2. If a system uses FIFO policy for page replacement, and it has 4-page frames with no pages loaded to begin with. Also, the system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order, then how many page faults will occur? 5
3. What is fragmentation? How it can be overcome? 5

2<sup>nd</sup> Internal Assessment Examination (2024)

Dept: CSE Semester: 6<sup>th</sup> Sub: Artificial Intelligence (PEC-CS502C)

4. What is reasoning? Explain different types of reasoning with examples. 1+4
5. Convert the following proposition into FOL : (a) Every girl walks or talks (b) Not all students like both maths or Science (c) There is a country that borders both India and Nepal 6
6. Consider the game play tree given below and apply alpha-beta pruning algorithm on it. 4



*Whiteboard*

## 1.2<sup>nd</sup> Internal Assessment Examination (2024)

Dept: CSE Semester: 6<sup>th</sup> Sub: Software Engineering (PCC CS501)

1. Define Data Dictionary and its significance. Explain Cohesion and Coupling in respect of good software design. 2+3
2. Explain different types of Software Maintenance. How we can calculate Maintenance Cost. 3+2
3. Write down the difference between Code Walkthrough and Code Inspection. Draw Control Flow graph for the following code and calculate cyclomatic complexity. 2+3

```
if A = 10 then
```

```
  if B > C
```

```
    A = B
```

```
  else A = C
```

```
endif
```

```
endif
```

```
print A, B, C
```

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE  
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
ODD SEM 2024

Paper Name: Operating System

Paper code: PCC-CS504

Time: 40 minutes

Full Marks: 15

Class Test: 1<sup>ST</sup>

Date: 28/09/2024

Answer any three questions.

1. "Context switching is a pure overhead to the system" -Justify the statement. What are the necessary and sufficient conditions for deadlock? *life cycle* (2+3)
2. What is indefinite postponement? What is the solution of it? If a computer has six tape drives, with n processes competing for them and each process may need two drives, what is the maximum value of n for the system to be deadlock free? (2 + 1 + 2)
3. What is semaphore? Give a solution of second reader-writers problem with semaphore. (1+4)
4. Write a monitor code to allocate a single resource and explain it. (5)

~~Item} → Price}~~

~~Address, item, Price) and F={S\_Name → Address, {S\_Name,~~  
Explain Different Anomalies that can arise while designing relations in a Database.

1<sup>st</sup> INTERNAL/ SOFT COMPUTING/ PEC-CS501B / 5<sup>TH</sup> SEM-CSE/30.11.2023/ FM: 15

1. What is Fuzzy set? Also define fuzzy membership function? 2+2= 4
2. Write the difference between Fuzzyness vs. Probability and Fuzzy Set vs. Crisp Set 2+2=4
3. What is activation function? Discuss about Bipolar Sigmoid function and Hyperbolic Tangent Function. 2+2=4
4. Classify the OR function truth table with single layer perceptron. 3

DBMS(PCC-CS503)/CSE / Internal-1 / 2024 Full Marks: 15/ Time: 45 Minutes

Answer any three questions: (3 X 5 Marks = 15 Marks)

1) What is Foreign key? Describe the concept of specialization and generalization in context of E-R data model. What do you mean by data independence? 1+3+1

2) i) Explain in brief 3-schema architecture of DBMS. 2+3

ii) Let the following relation schemas be given and write the SQL queries:

EMPLOYEE(EMP\_no, E\_name, E\_address, E\_ph\_no, Dept\_no,  
Dept\_name, Job\_id, Salary)

Department(Dept\_no, Dept\_name, Manager\_id, Address)

a) Display all the dept numbers available in emp and not in dept tables and vice versa.

b) Display total salary spent for each job category.

c) Display lowest paid employee details under each manager.

3) Find the normal form of the following relation with the Functional Dependency Set F, 5  
Supplier(S\_Name, Address, Item, Price) and  $F = \{S\_Name \rightarrow Address, \{S\_Name, Item\} \rightarrow Price\}$

4) Explain Different Anomalies that can arise while designing relations in a Database. 5

1<sup>st</sup> Internal Assessment Examination (2024)

Dept: CSE Semester: 6<sup>th</sup>

Sub: Software Engineering (PCC CS501)

(Answer any three)

1. What is prototype? Write down the advantages and disadvantages of prototyping model. Write down the differences between Evolutionary Model and Prototyping Model. 1+3+1
2. What do you understand by the term phase containment of errors? Why it is important? Discuss Spiral Model with suitable diagram. 1+1+3
3. Assume that you want to develop a software product of real time system. And that its size is 50,000 lines of code. Assume that the Average salary of software developers is Rs. 40,000 per month. Determine the effort required to develop the software product, the nominal development time, and cost to develop the product. 2+2+1
4. What is the significance of SRS Document? Draw Decision Table for Library Automation Software. 2+3

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE

JGEC/B.TECH./CSE /PCC-C3502/2024

DISCRETE MATHEMATICS (FIRST CLASS TEST)

Full Marks: 15

Time: 45 Minutes

Answer any *three* questions

3x5=15

1. i) Find  $f \circ g$  where  $f: \mathbb{Z} \rightarrow \mathbb{Z}$  is defined by  $f(n) = n^2, n \in \mathbb{Z}$  and  $g: \mathbb{Z} \rightarrow \mathbb{Z}$  is defined by  $g(n) = 2n, n \in \mathbb{Z}$

2

- ii) Show that the two statements  $(p \wedge q) \rightarrow r$  and  $(p \rightarrow r) \wedge (q \rightarrow r)$  are not logically equivalent.

3

2. Find  $f \circ g, g \circ f, f^{-1}$  and  $g^{-1}$  where

5

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 3 & 5 & 1 & 6 \end{pmatrix}, g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 6 & 4 & 5 & 3 & 2 \end{pmatrix}$$

3. i) Suppose that  $A \times B = \emptyset$ , where  $A$  and  $B$  are sets, what can you conclude?

$b^2c$

2

- ii) Construct the truth table for the following proposition

3

$$(p \rightarrow q) \rightarrow (q \rightarrow p)$$

4. A relation  $\rho$  is defined on the set  $\mathbb{Z}$  by  $a \rho b$  if and only if  $a b > 0$ , for  $a, b \in \mathbb{Z}$ .  
Examine if  $\rho$  is i) reflexive, ii) symmetric, iii) transitive

5

5. Show that the given mapping  $f$  is bijection. Determine  $f^{-1}$ .

5

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } f(x) = 2x + 3, x \in \mathbb{R}.$$

2.1